

R1

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1 New Code / Devs

1.1 Python Terminal Script for Filtering

Because we're likely to need to filter quite a few systems with very similar methods I decided to make an actual script instead of just using a notebook. The largest upside is that this can easily be run on a compute node, which should make filtering larger sims easier.

Takes a user input in the CLI (see below), filter for all BH_Sol systems, and returns the population file with said system as well as a .csv of just the BH_Sol rows. Furthermore, as it is running it provides an output via terminal stating how many BH_MS systems it found, how many are S1:S2, as if any are S2:S1, it prints a warning as well as listing the count of them.

Usage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
-o	output location	.results
-outputName	name for the outputed files ¹	BH_Sol_Systems
-overwrite	Whether to overwrite existing results, T/F bool	True
--maxMass	Max mass used when filtering, defaults to $3M_{\odot}$	3.5
--maxPeriod	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

¹If a name is not specified, it will use the original filename, and append a specifier to the front

2 New Research

When filtering via notebook and script, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, that being ZAMS_oCE1_CC1_oRL02_CC2. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then post-MS **unstable** RLO happens via $S1 \rightarrow S2$. This leads to common envelope, which *greatly* decreases the orbital period. This envelope is then ejected, leaving a depleted stripped He core. The core then collapses in a couple thousand years, leaving the system in a BH-MS state with low p_{orb} /seperation.

These systems then evolve in a semi-peculiar way, where it is highly likely that there will be RLO from S2 onto S1, and because $S1 \nless S2$, this is stable, and leads to *very* prolonged stable RLO (seemingly on the scale of $> 14Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestingly, the initial period for all the systems is very close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the initial masses of S1 and S2 appear to be quite constrained as well.

	orbital_period.i
count	6.000000
mean	4022.716197
std	151.522384
min	3873.118288
25%	3908.300983
50%	3978.305326
75%	4124.397101
max	4248.507680

	S1_mass.i
count	6.000000
mean	17.515761
std	0.520269
min	16.815111
25%	17.190036
50%	17.510176
75%	17.818155
max	18.254965

	S2_mass.i
count	6.000000
mean	2.296788
std	0.320702
min	1.797809
25%	2.120753
50%	2.353828
75%	2.547210
max	2.620529